

1. Raw Sensor Data

CMOS Array
Photon-to-Voltage

Bayer CFA Filter
 $1R : 2G : 1B$ mosaic

2. Pre-Processing

Demosaicing
Bilinear Interpolation

RGB $\rightarrow$ HSV
Isolate Chrominance

Binary Threshold
 $H_{\min} \leq H \leq H_{\max}$

3. Feature Extraction

Blob Detection
Connected Components

Area Filtering
 $\sum \text{pixels} > \text{noise}$

Image Moments
 $\bar{x} = \frac{M_{10}}{M_{00}}, \bar{y} = \frac{M_{01}}{M_{00}}$

4. Control Logic

Error Calculation
 $e(t) = \bar{x} - X_{\text{center}}$

PI Controller
 $u(t) = K_p e + K_i \int e dt$

PWM Motor Commands
 $v_{\text{left}}, v_{\text{right}}$